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The AI-GPR Index: Measuring Geopolitical Risk using Artificial Intelligence

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Measuring geopolitical risk: A new tool

- Economists have long tried to quantify hard-to-measure phenomena by counting specific words or combinations of words in newspapers — fast, transparent, and powerful. E.g. the BBD's EPU Index, or GPR Index of Caldara-Iacoviello (2022)
- Limitation of these approaches: dictionary-based approach cannot easily tell whether "war" in an article means active conflict or retrospective on WWII; cannot say who is threatening whom; cannot tell whether crisis threatens oil supplies.
- Large language models change this: LLM reads each article and understands its meaning — just as a research assistant would, but at a scale that is impossible for humans.

What this opens up

Better measurement of the same concept (less noise, fewer false positives) and new measures that word-counting simply cannot produce: anticipated threats vs. realized acts, oil-specific risk, directed bilateral stress

What we do

We build a daily measure of geopolitical risk from 4.6 million newspaper articles spanning 1960 to present. Each article is read and scored by an AI model (GPT-4o mini).

We extend the same approach with additional classification layers to produce new measures that keyword-based methods cannot produce.

1. The AI-GPR Index

- Uses definition of geopolitical risk from Caldara and Iacoviello (2022)
- LLM scores every article 0–1 for geopolitical risk intensity
- Available daily from 1960 onwards

2. AI-GPR Measures

- **Threats and Acts**: is the risk anticipated (threat) or realized (act)?
- **Oil-GPR**: does this article discuss a geopolitical oil supply disruption, and if so where?
- **Bilateral GPR**: who is acting on whom: initiator, respondent, spillover countries

Building the index: Step 1

Collecting Articles

Cast a **very** wide net to collect every article *possibly* discussing geopolitical risk

Article query:

```
airstrike* OR alliance OR annex* OR attack* OR
blockade* OR bomb* OR cease-fire OR combat OR
conflict* OR coup OR crisis OR diplomat* OR embargo
OR enem* OR hostage* OR hostil* OR invade* OR
invasion* OR militia OR military OR missile*
OR nuclear OR refugee* OR riot OR sanction* OR
sovereign OR terror* OR treaty OR troop* OR truce
OR unrest OR violence OR war OR weapon* ...
```

Numbers:

- 4.6 million pass the article query filter
- ~8.5 million total articles (NYT, WaPo, Chicago Tribune, 1960–2025)
- Filter is very conservative by design; false positives handled in Step 2

Step 2: Full classification prompt (simplified)

Pass the articles to LLM model, and evaluate its GPR content

```
You will be given [the first 2,000 characters of] a news article.
Classify the article's assessment of geopolitical risk based only on what the article states or
strongly implies.
Geopolitical risk is defined as the threat, realization, and escalation of adverse events associated
with: wars (...); escalation of existing wars (...); major terrorist attacks (...); and tensions
among states and political actors (...).

Assign a geopolitical risk score from 0.0 to 1.0 based on the following scale:
0.0--0.2: No mention of geopolitical risks.
0.2--0.4: Mentions of minor tensions, diplomatic disputes, or isolated incidents....
0.4--0.6: Discussion of significant tensions...
0.6--0.8: Substantial discussion of major war initiation/escalation risks....
0.8--1.0: Extensive coverage of imminent or new war, major war escalation, severe terrorism threats,
or critical threat to international stability.

Movies, books, anniversaries of old events, and obituaries should receive a score of 0.0 ...
Purely domestic political events (elections, protests, internal policy debates) should score 0.0 ...
```

The scoring scale in practice: Results

Score: 0.2 (minor tensions)

Kim Jong Un takes his daughter to China — and fuels succession chatter

Kim Ju Ae, who is about 12, has accompanied her father on a high-profile visit to Beijing, making it more likely that she is North Korea's heir apparent.

September 3, 2025

Score: 1.0 (imminent or active war / severe threat)

The Legality, or Illegality, of Killing a Foreign Leader, Explained

The United States and Israel started a war with Iran by killing its supreme leader, Ayatollah Ali Khamenei.

Listen · 8:48 min

Share full article

Calculating the AI-GPR index

Daily index formula

$$\text{AI-GPR}_t = \sum_{i=1}^{N_t} S_{it} \times \frac{1}{A_t} \times \frac{1}{\bar{S}}$$

- S_{it} = GPR score assigned by LLM to article i on date t
- N_t = number of scored articles on date t (those passing the filter)
- A_t = Total articles published on date t (denominator, not just filtered)
- $\bar{S}$ = normalization constant

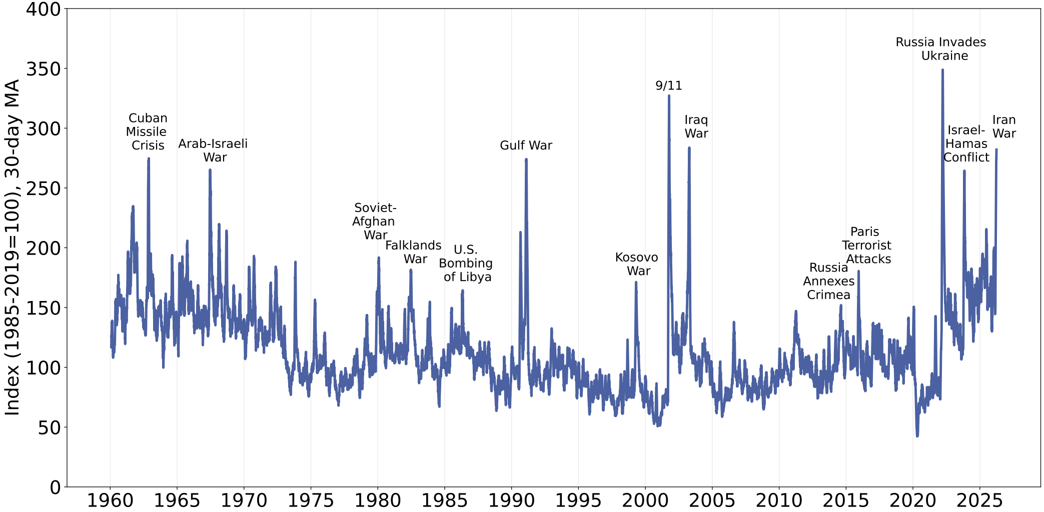
Result of Steps 1+2:

- 4.6M articles filtered
- 1.2M (26%) receive score > 0
- 14% of total articles are GPR-relevant

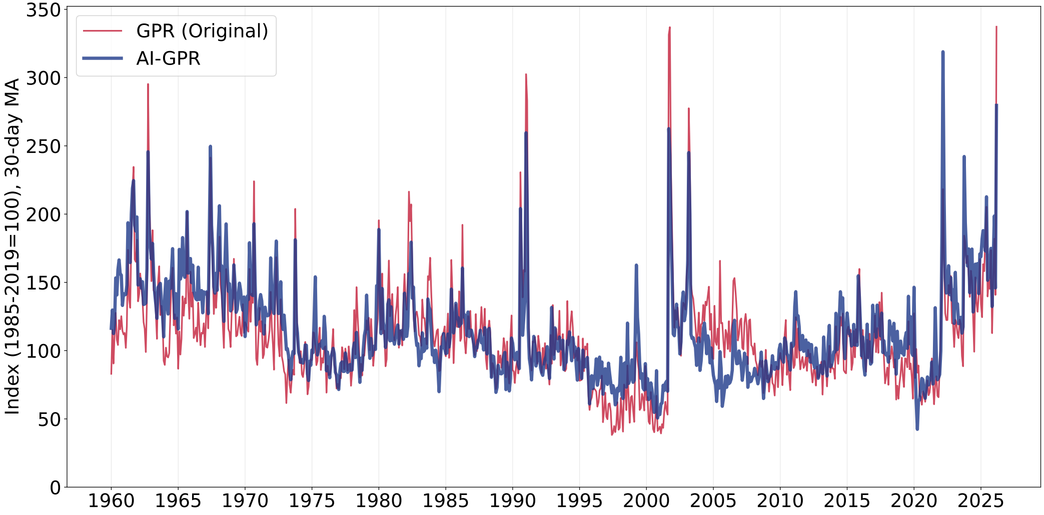
Key properties:

- Daily frequency, 1960–2026
- Continuous intensity (not binary)
- Correlation 0.69 with original GPR; higher persistence (autocorr. 0.73 vs. 0.62)

The AI-GPR index, 1960–2026



AI-GPR vs. original GPR: Monthly comparison (1960-2026)



How accurate is the AI-GPR? Russia's invasion of Ukraine (2022)

Daily AI-GPR (blue) and keyword-based GPR (red) around Russia's invasion of Ukraine. The AI-GPR spikes sharply on invasion day with less day-to-day noise.

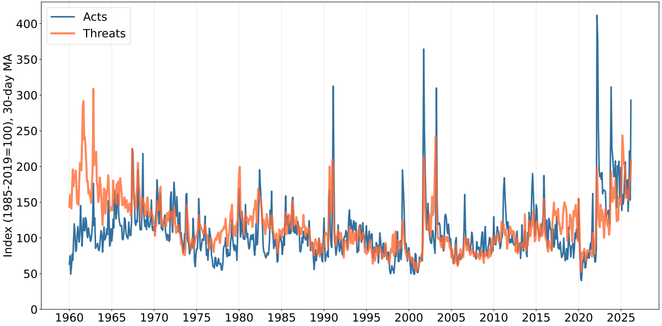
Measurement validation

Concern	Test	Finding
Stochasticity	Re-run 3x on 1,050 articles	Correlation 0.99 across runs
Text truncation	2,000-char vs. full text	Correlation 0.93; 87% identical scores
Bias vs. humans	Human audit (˜2,000 articles, 3 coders)	Correlation with human consensus: 0.85 Correlation between humans: 0.78; False-discovery rate 13% (Keyword-based:16%) Miss rate 17% (Keyword-based: 81%)
Model sensitivity	GPT-4o, GPT-5, Llama 3.1	Cross-model correlation 0.84–0.94
Hindsight bias	LLM with twisted prompt: "Your knowledge cutoff is today!"	Correlation with baseline 0.95
Type-II error	LLM on excluded articles	$<$1% receive positive score

New measures: Threats vs. acts

A second classification layer

Each article with a positive GPR score is re-read and labeled a **threat** (anticipated risk), an **act** (event occured or underway), or **both** (∼15%, counted in each series). Subindexes built exactly like the headline index.



Application 1: Geopolitical risk and stock returns

	(1) OLS	(2) Decomp.
ΔGPR_t	-0.136^{**}	
	(0.057)	
Persistent $\widehat{\Delta\text{GPR}}_t$		-0.271^{**}
		(0.134)
Shock $\hat{u}_t$		-0.119^{*}
		(0.062)
Obs.	3,452	3,448
Sample	1960–2026	

Weekly excess returns (%). Pairs-bootstrap SE in col. (2).

GPR and stock returns (col.1)

- Analyze relationship between changes in GPR and changes in weekly stock returns
- OLS coefficient is negative and significant
- Same exercise with keyword-based GPR: OLS coefficient imprecisely estimated $\Rightarrow$ AI-GPR provides a sharper signal

Decomposing stock returns (col.2)

- Decompose ΔGPR into:
  - persistent: AR(4) fitted values
  - shock (residuals) component
- Persistent component coefficient (-0.27) is twice aggregate OLS (-0.14):
- Markets price "long tail" of geopolitical risk, not just short-term movements

Threats vs. acts: Priced very differently

	(1) OLS	(2) Decomp.
$\Delta GPR_{threats}$	-0.081^{**}	
	(0.037)	
ΔGPR_{acts}	-0.084	
	(0.061)	
Persistent, threats		-0.220^{**}
		(0.103)
Persistent, acts		-0.063
		(0.131)
Shock, threats		-0.052
		(0.044)
Shock, acts		-0.090^{**}
		(0.042)

Weekly excess returns (%), 1960–2026.

In levels (col. 1)

- Threats: precisely estimated negative effect
- Acts: similar size, but too noisy to register

Decomposed (col. 2)

- Threats bite through their **predictable** part: markets re-price gradually as a threat persists
- Acts bite through their **surprise** part: foreseeable events are already in prices

Takeaway

What matters is not the level of geopolitical risk but whether it is anticipated or unexpected — a distinction a single regression cannot capture

New measures: Geopolitical oil supply disruptions

Two-stage LLM classification

Stage 1: every article receives a GPR score 0–1 (AI-GPR)
Stage 2: articles with GPR > 0.5 that contain oil keywords are re-classified: does this article discuss an *oil supply disruption caused by a geopolitical event*? If yes, which region?

Coverage (of articles with GPR > 0.5)

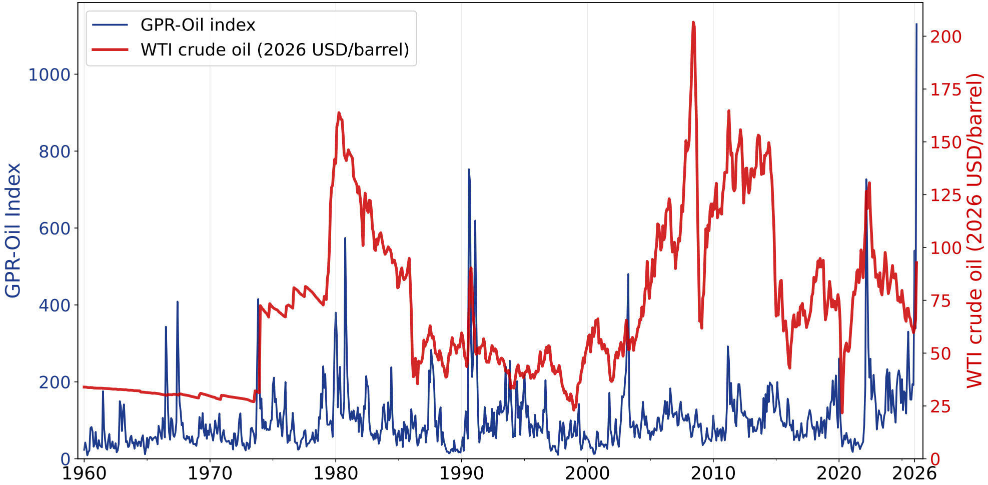
- 12.6% contain oil keywords
- 9.3% classified as oil disruption

Regional breakdown

- Middle East: 63%
- Russia: 14% (surges after 2014)
- North Africa: 11%
- Venezuela, W. Africa, SE Asia: 5–6% each

$\text{Oil-GPR}_t = \sum_{i \in \text{disruptions}} S_{it} \,/\, A_t$ (weighted by GPR intensity, normalized by total articles)

OIL-GPR index and oil prices



Oil-GPR index (left axis, 30-day MA) and WTI oil price (monthly, right axis). Sample ends March 31, 2026

Application 2: A proxy SVAR with geopolitical oil price shocks

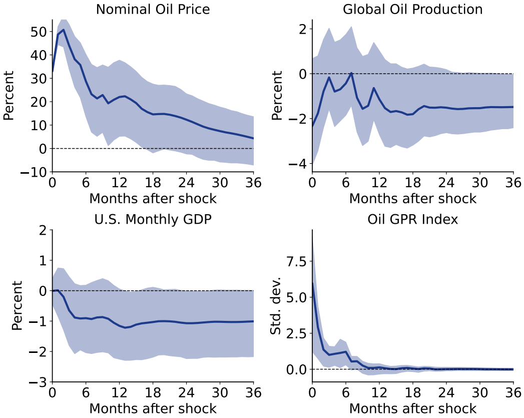
External instrument

- We use cumulative monthly WTI log-price changes on days when Oil-GPR spikes > 2 s.d. above its mean to construct an external instrument for a VAR.
- These are days with a clear geopolitical trigger—not OPEC quota decisions, demand fluctuations, or weather events.
- Oil-GPR spikes provide a valid instrument: correlated with geopolitical supply shocks, uncorrelated with demand shocks.

VAR setup

We embed this variable in a proxy SVAR with: real oil price (WTI), global oil production, U.S. real GDP, and the Oil-GPR index. 12 lags, monthly, 1975–2024.

Proxy SVAR: Impulse responses to a geopolitical oil shock



Impulse responses to a one-standard-deviation geopolitical oil supply shock (normalized to 33% oil price rise on impact). Shaded bands: 80% bootstrap CI. 12 lags. Sample: 1975–2024.

New measures: Who acts on whom? The actor-role classifier

```
TASK: Identify the key country-level actors in this geopolitical event and assign each a role.

Roles are defined as follows:
- initiator:   the country that launched or triggered the hostile action (military attack, invasion, sanctions, blockade, etc.)
- respondent:  the country that is the direct object of the hostile action
- spillover:    a country not directly involved but significantly affected (energy shock, refugee flows, trade disruption, etc.)

Rules:
- List up to 5 actors total; include only countries with a meaningful role in THIS article
- Use standard country names (e.g.  ‘‘United States’’, ‘‘Russia’’, ‘‘Saudi Arabia’’)
- Assign exactly one role per actor
- When initiation is genuinely contested (civil wars, mutual escalations), assign both parties the role of initiator
- If the article is too vague to identify actors, return an empty actors list
```

Applied to all articles with GPR score > 0. Enables country-level GPR indices (200 countries on the companion webpage) and directed bilateral series for the top 1,200 country pairs.

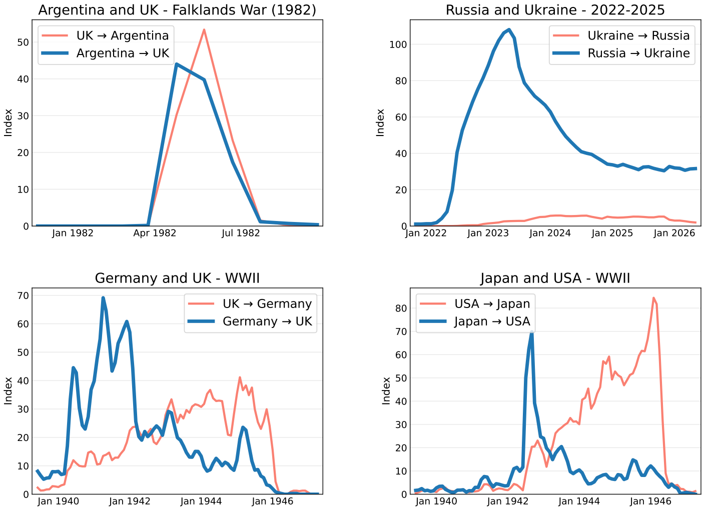
Bilateral geopolitical risk: A directed measure

What it Measures

For each country pair (i, j): the share of articles in month t where country i is the *initiator* and country j is the *respondent*.

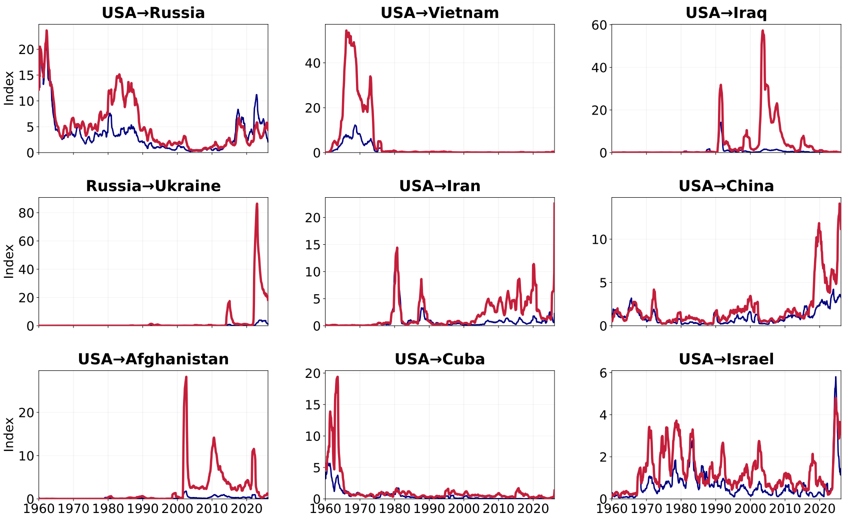
Directed : Russia acting on Ukraine $\neq$ Ukraine acting on Russia.

- Constructed for 1,200 country pairs
- Extends CI2022 country indexes to a full directed bilateral matrix



Selected pairs by bilateral GPR

Most recurring bilateral GPR indexes

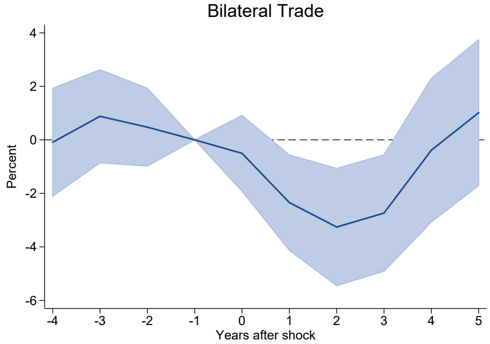


Application 3. Does bilateral GPR depress bilateral trade?

Local Projection

$$y_{od,t+h} - y_{od,t-1} = \gamma_h Z_{od,t} + \alpha_{od,h} + \alpha_{o,t,h} + \alpha_{d,t,h} + \mathbf{d}_h$$

- $y_{od,t} = 100 \times \log$ two-way trade, so γ_h is a percent response.
- $Z_{od,t}$: bilateral AI-GPR, standardized within pair.
- Pair FE, origin$\times$year and destination$\times$year FE.
- Controls: lags of trade growth and of GPR, trade agreement.



Percent response of total two-way bilateral trade to a one-standard-deviation rise in bilateral GPR, relative to year $t-1$. Shaded area is the 95% confidence interval, standard errors clustered by pair. Comtrade, 1962–2020.

Result

Conclusions

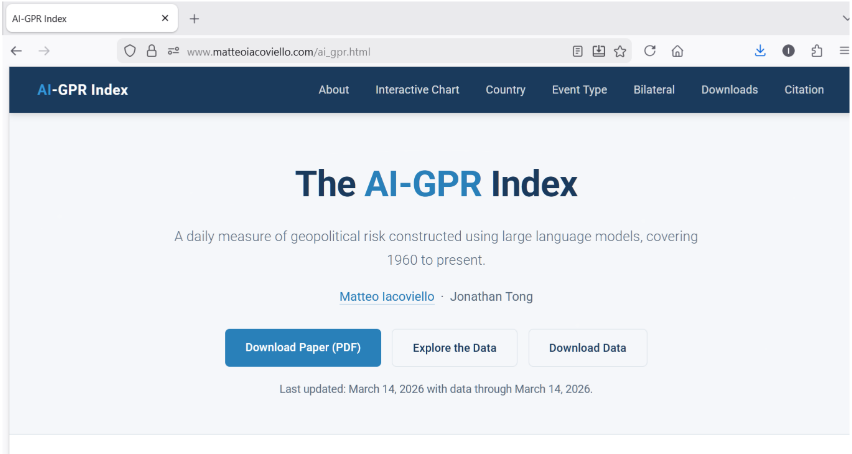
The AI-GPR Index

We construct an LLM-based measure of geopolitical risk with greater precision than keyword methods, and can measure GPR-related phenomena that cannot be easily measured otherwise

Three Applications

- **Stock returns**: AI-GPR delivers a sharper negative effect; markets are especially sensitive to the persistent component of GPR — threats are priced when anticipated, acts only when unanticipated
- **Oil shocks**: two-stage LLM identifies 65 years of geopolitical oil disruptions; proxy SVAR shows adverse effects of GPR-related oil price shocks
- **Bilateral trade**: directed country-pair indices (who acts on whom) — bilateral GPR predicts bilateral trade declines

All data available at `matteoiacoviello.com/ai_gpr.html`



Daily and monthly indices, country decompositions, bilateral pairs, event types — all downloadable. Interactive charts with smoothing controls.